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CE205
Databases and Information Retrieval

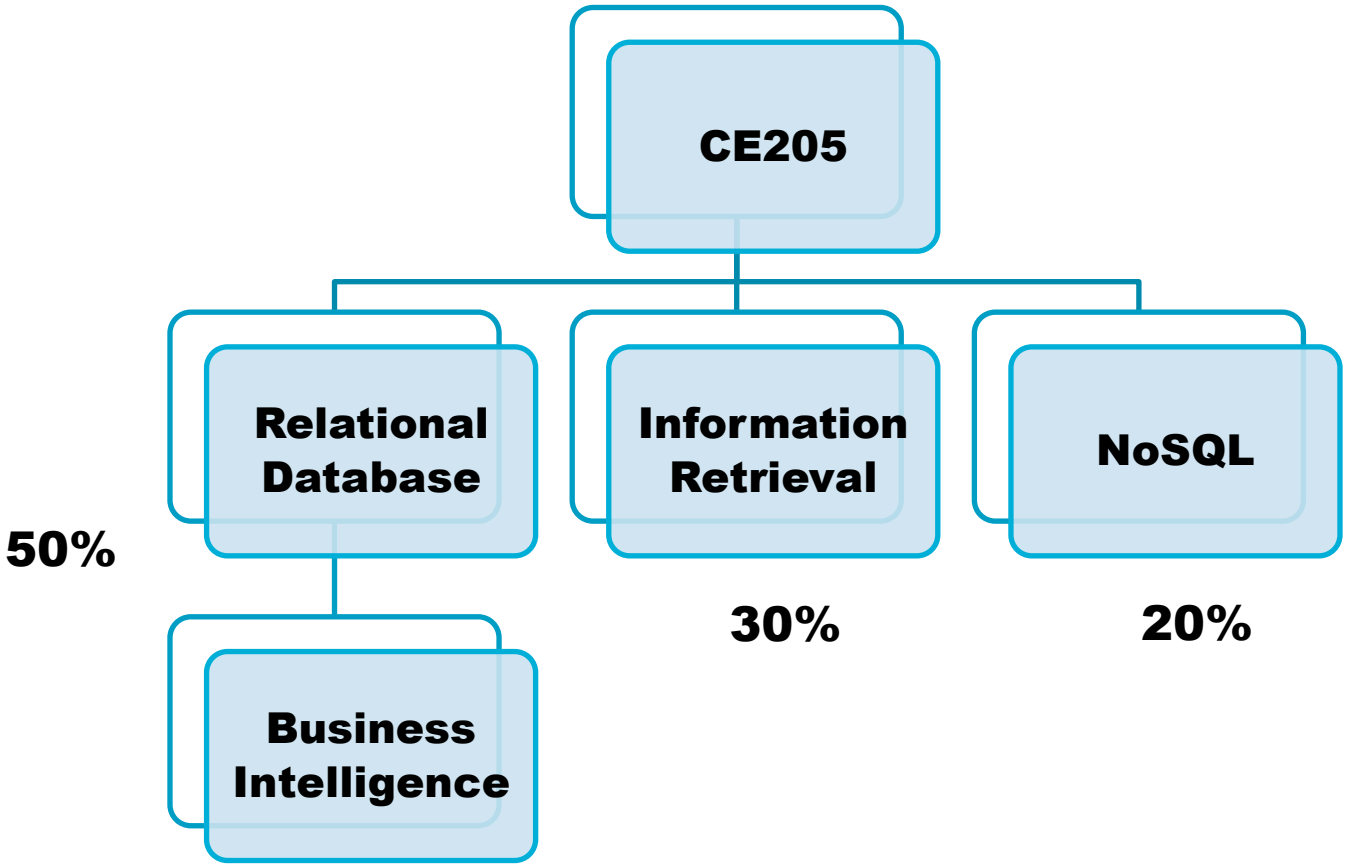
Module Information

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Lecturer
School of Computer Science and Electronic Engineering

Outline

- Content Distribution in CE205
- Syllabus
- Lectures/Classes/Labs
- Assessment
- Books and Additional Resources

Content Distribution in CE205



(Questions in the final exam will follow the same distribution)

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Syllabus

- SQL Database Design Principles
  - Modelling in a Realistic Scenario
  - SQL Database Design Using Normalisation
  - Operational/transactional Systems and Business Intelligence
  - Concepts and Design of Data Warehouses
- Principles of Information Retrieval
  - Document Processing and Indexing
  - Term-document Matrix and Bag-of-Words Model
  - Term Weighting and Information Retrieval (IR) Models
  - Performance Evaluation in a Practical Task and Relevance Judgments
- NoSQL Databases
  - Introduction of NoSQL Databases and Working with MongoDB
  - Comparison of SQL, IR and NoSQL paradigms

Timetable

- **Two-hour lecture each week:**
  - **Monday 09:00 - 11:00 (weeks 2-11), Room: EBS.2.34**
- **One-hour class each week:**
  - **Wednesday 11:00 - 12:00 (weeks 2-11), Room: LTB02**
- **One-hour lab each week :**
  - **Monday 13:00 - 14:00 (weeks 3-5, 7, 9-10), Lab: CSEE Lab07**

(always keep an eye on your timetable)

Lectures

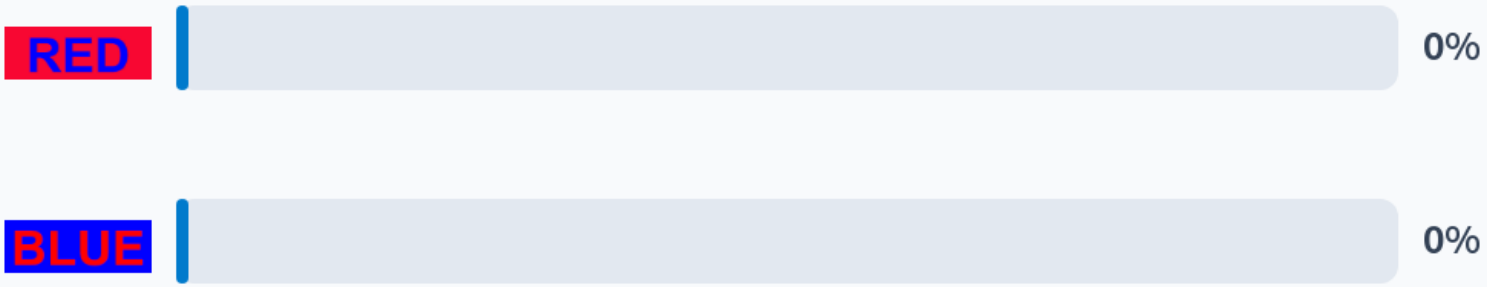
- **Slides/handouts for each lecture/class will be available electronically on Moodle well before the lecture/class day.**

- **For interactive learning and formative assessment, I will use Poll Everywhere with the link pollev.com/zali085** *(copy and paste in the browser)*

(Students are strongly advised to attend the lectures)

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Please, select blue option.



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Classes/Labs

- **In the classes (1h/week) we will:**
  - **solve the exercises/problems together**
- **In the labs (1h/week) we will use:**
  - **MySQL Workbench to implement relational databases**
  - **MATLAB for data analysis/machine learning in business intelligence**
  - **MongoDB (with Python) for NoSQL**

(you will not be assessed based on the programming skills in the final exam)

Assessment

- **Examination**
  - **Two-hour final exam worth 70% during Summer Examination Period.**
- **Assignments**
  - **Assignment 1: worth 10%**
  - **Assignment 2: worth 10%**
- **Progress Test (multiple-choice questions)**
  - **worth 10%**

(Check submission deadlines on Faser)

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“What do I need to know?”

- **How much mathematics do I need to know?**
  - **Most of the mathematics will be of basic level set theory and calculus. I will give an overview, when required.**
- **How much programming is there in the course?**
  - **This is not a programming course, and you will not be assessed based on the programming skills in the final exam.**
- **Do you need the prior knowledge of CE153: Introduction to Databases?**
  - **Some maturity in database design is required. However, I will make the lecture independent to CE153 and will not consider any prior knowledge.**

When poll is active, respond at **pollev.com/zali085**

Text **ZALI085** to **07480 781235** once to join

Did you bunk the class in high school when "Functions f(x)" were being discussed?

Yes

No

Powered by Poll Everywhere

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Office Hours

- **Monday 11:00 – 13:00**
- **Weeks: 2-11**

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Books

- **Relational Databases**

  D1: Database processing: fundamentals, design, and implementation *(David M. Kroenke; David J. Auer) . 14th Edition (Global Edition). Pearson.*

  D2: Database Systems: A Practical Approach to Design, Implementation and Management *(Connolly, T. M., & Begg, C.E.). 6th Edition. Addison Wesley.*

- **Information Retrieval**

  R1: Information Retrieval: Concepts, Models and Systems *in Computational Analysis and Understanding of Natural Languages: Principles, Methods and Applications (Chapter by Venkat N. Gudivada; Dhana L. Rao; Amogh R. Gudivada). Elsevier.*

  R2: Introduction to Information Retrieval *(Manning, C. D., Raghavan, P., & Schütze, H. (2008). Cambridge University Press. Available online at https://nlp.stanford.edu/IR-book/*

Books *(continue)*

- **NoSQL**
  - **N1: NoSQL with MongoDB in 24 Hours (Brad Dayley*), Pearson Education.***
    - **Hour 16: Implementing MongoDB in Python Applications**

Additional Resources

- A very good online chapter (book1 of Relational Databases) for Managing Databases with MySQL 5.7 (Online Chapter 10C)
  - Go to the link (Pearson - Online Resources for Kroenke (pearsoncmg.com)and click on online chapters (of Database Processing, 15/e, David Kroenke • David Auer), three chapters 10A, 10B and 10C will be downloaded
  - You can also download ”Students Data Files”. These files are databases implemented in MySQL. The details of these databases is given in the book (D1).
- Other resources will be provided with the lectures.